

Software Engineer | Full-Stack | AI / LLM

Ayomikun Festus-Olaleye

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Professional Summary

Full-stack Software Engineer with 5+ years shipping production systems across the complete stack: React/TypeScript frontends, Python/FastAPI backends, Docker containerisation, GitHub Actions CI/CD, and LLM orchestration pipelines. Engineered a fault-isolated asyncio backend orchestrating 4 concurrent Claude + GPT-4 analysis pipelines with 80% Pytest-gated test coverage and mypy strict enforced on every pull request. Raised a live booking platform from Lighthouse 62 to 91 (+47%) and cut API calls by 60%. MSc Software Engineering, Blekinge Institute of Technology.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Python, Java

Frontend: React.js, Angular, Redux, HTML5, CSS3, Tailwind CSS, Shadcn UI, Micro Frontend Architecture, Responsive Design

Backend & APIs: Node.js, Python/FastAPI, Pydantic v2, RESTful API Design & Integration, GraphQL (familiar), Java/Spring Boot (familiar)

Databases: MongoDB, PostgreSQL, MySQL, SQL Server

DevOps & Cloud: Docker, GitHub Actions CI/CD, Git, Netlify, Azure Cloud Platform (familiar), Kubernetes (familiar), Terraform (familiar)

Testing & QA: Jest, Cypress, Pytest, Vitest, End-to-End Testing, mypy Strict Mode, TypeScript Strict Mode (zero any)

Engineering: Microservices Architecture, Clean Architecture, Performance Optimisation, Code Review & Mentoring, Agile/Scrum, GDPR & WCAG Accessibility

AI & ML: LLM Orchestration, Claude API, OpenAI API (GPT-4), Prompt Engineering

Experience

Software Engineer

Dec 2023 – Aug 2024

Financial, Forensic Studies & Diagnostic Ltd (FFSD) • Lagos, Nigeria

ffsdtravels.com

- Architected and delivered **FFSD Travels** end-to-end component architecture, Redux state design, typed REST API service layer, Docker containerisation, and Netlify CI/CD serving real users in production.
- Raised Lighthouse performance score from **62 to 91 (+47%)** via lazy loading, route-based code splitting, API caching, and memoisation across a microservices-integrated React/TypeScript frontend.
- Reduced production API calls by **60%** by introducing debounced search and request-level caching in Redux, measured via network profiling.
- Integrated **5+ third-party REST APIs** (flights, hotels, payments) through a service-layer abstraction that fully decouples UI components from data-fetching concerns.
- Grew frontend test coverage from **0% to 70%** by authoring Jest unit tests and Cypress end-to-end tests across all critical booking, search, and checkout flows.
- Eliminated UI regression bugs zero design-regression QA reports in the final two sprints by engineering a reusable library of **30+ Shadcn UI / Tailwind CSS components** as the single source of truth across the codebase.
- Standardised deployments with **Docker** and **Netlify CI/CD**, eliminating environment drift and ensuring reproducible builds across dev, staging, and production.
- Ensured **GDPR compliance** and **WCAG accessibility** across all customer-facing flows; coached junior developers via code review and pair programming on architecture, testing, and accessibility standards.

Software Engineer

Jan 2023 – Nov 2023

Pedistack • Fintech • Remote

- Eliminated stale-state and race-condition defects by implementing **Redux state management** with structured action patterns and typed REST API integrations.
- Reduced reported UI layout bugs by **25%** (by QA ticket volume) via consistent responsive design across desktop and mobile breakpoints.
- Cut code duplication across **4+ product areas** by building a shared library of 20+ modular React components, establishing a single source of truth for UI logic across the team.

Independent Software Engineer

Oct 2021 – Dec 2022

Self-Directed Engineering Projects • Remote

- Built **MikunStore**, a full-featured e-commerce application with multi-category browsing, cart, checkout, and client-side PDF receipts validated with Cypress end-to-end tests across the full purchase funnel (React 18, Vite, Context API, jsPDF, Cypress, Tailwind CSS).
- Developed **CineVault**, a type-safe movie discovery platform consuming the TMDB REST API with genre filtering and per-request language headers; TypeScript strict mode enforced across all API shapes (React 18, TypeScript, Redux Toolkit, Axios, Tailwind CSS).
- Shipped a multilingual **Portfolio** with working contact form, i18n support, and a Vitest unit test suite; deployed to production via Netlify CI/CD (React 18, TypeScript, Vite, Vitest, EmailJS).

Software Engineer Intern

Apr 2021 – Sep 2021

PricewaterhouseCoopers (PwC) • Lagos, Nigeria

- Delivered enterprise **Angular** UI components to PwC's compliance and accessibility standards (QA sign-off within sprint timelines) by collaborating across product, QA, and backend teams in a structured SDLC.
- Contributed to UI testing cycles and technical documentation, building foundational knowledge of enterprise-grade software delivery at scale.

Projects

ReqSmell AI Requirements Analysis Tool (MSc Thesis)

2025 – 2026

React 18 · TypeScript · Redux · Python · FastAPI · Claude API · GPT-4 · Pydantic v2 · Pytest · GitHub Actions · [reqsmell.netlify.app](#)

- Built a production full-stack AI research tool as two independently deployed services: a React/TypeScript SPA frontend and a Python/FastAPI backend with a Node.js build pipeline enforcing TypeScript strict mode (zero **any**) and Ruff linting end-to-end.
- Engineered **4 concurrent async LLM pipelines** (Claude API + GPT-4, ambiguity + inconsistency detection) via asyncio orchestration with fault-isolated execution, zero cross-pipeline contamination during thesis evaluation.
- Enforced **80 % backend coverage and zero type errors on every PR** via GitHub Actions CI: Pytest, mypy strict, ruff linting, and TypeScript strict mode gates.
- Delivered **client-side PDF report generation** (multi-page comparison reports with statistics and charts) using jsPDF + html2canvas no server-side rendering required.

MikunAir Full-Stack Flight Booking Application (Portfolio)

2025 – 2026

React 18 · TypeScript · Apollo Client · Node.js · Express · GraphQL · Drizzle ORM · PostgreSQL · Zod · Vitest · Playwright · Docker · GitHub Actions · [MikunAir](#)

- Built a production-grade full-stack booking system as two independently containerised services a React 18/TypeScript SPA (Vite, Tailwind, Apollo Client) and a Node.js/Express/TypeScript API, each with its own GitHub Actions CI pipeline enforcing TypeScript strict mode and ESLint zero-warning gates.
- Implemented a **dual-protocol API**: GraphQL (Apollo Server, depth-limited) for flight search with rich **Airport** object types, and REST (Express + Zod validation) for bookings, authentication, and profiles, all responses projected through explicit DTOs.
- Engineered **overbooking prevention** using PostgreSQL **SELECT FOR UPDATE** inside a Drizzle ORM transaction to serialise concurrent seat-decrement operations; an integration test fires two simultaneous requests against the last available seat and asserts exactly one succeeds.

Education

MSc Software Engineering

Sep 2024 – Jun 2026

Blekinge Institute of Technology, Sweden

Thesis: *AI-assisted Detection of Requirements Smells* | Courses: Software Architecture • Applied AI • Software Metrics • Software Quality

B.Tech. Computer Science

Sep 2018 – Aug 2022

Bell University of Technology, Nigeria

Second Class Honours | CGPA: 3.43 / 5.0

Certifications & Languages

Microsoft SQL Server (2022) • Java Programming (2022) • Microsoft Office Specialist (2022) • SQL Server Administrator (2020) | **English**: Professional Proficiency (C1)